CAMBRIDGE A2 LEVEL MATHEMATICS (9709) • UNIT 0.01

A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness

ANSWER KEY & SOLUTIONS

Pure 3 & Probability/Statistics 2 Readiness • Cambridge 9709 **Total Questions:** 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 D
Q6 A	Q7 A	Q8 C	Q9 C	Q10 A
Q11 D	Q12 B	Q13 A	Q14 A	Q15 B
Q16 D	Q17 D	Q18 C	Q19 A	Q20 A
Q21 A	Q22 D	Q23 D	Q24 D	Q25 B

A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness Practice Exercises

1 MCQ

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

A $\frac{2}{x-1}+\frac{3}{x+2}$ ✓

B $\frac{3}{x-1}+\frac{2}{x+2}$

C $\frac{2}{x-1}-\frac{3}{x+2}$

D $\frac{1}{x-1}+\frac{4}{x+2}$

WORKED SOLUTION & VERIFICATION

To decompose the rational function into partial fractions, set up the identity with distinct linear factors:

$$\frac{5x+1}{(x-1)(x+2)} \equiv \frac{A}{x-1}+\frac{B}{x+2}$$

Multiplying both sides by the common denominator $(x-1)(x+2)$:

$$5x+1 \equiv A(x+2)+B(x-1)$$

To determine A, substitute $x=1$:

$$5(1)+1=A(1+2)+B(0) \implies 6=3A \implies A=2$$

To determine B, substitute $x=-2$:

$$5(-2)+1=A(0)+B(-2-1) \implies -9=-3B \implies B=3$$

Therefore, the partial fraction decomposition is:

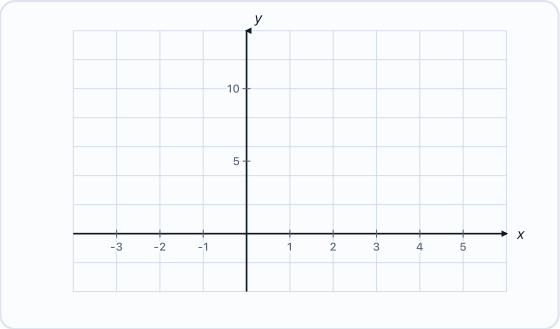
$$\frac{2}{x-1}+\frac{3}{x+2}$$

Common Misconceptions:

- Choice B incorrectly associates A with the root $x=-2$ and B with $x=1$, reversing the numerators.
- Choice C results from a sign error when solving $-9=-3B$.
- Choice D assumes coefficients must sum directly to 5 without properly weighting by denominator factors. [M1] [A1]

2 MCQ

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?



A 2 ✓

B 6

C $\frac{2}{3}$

D $\ln 3$

WORKED SOLUTION & VERIFICATION

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule.

Let $u = 2x^3 + 1$, so $y = \ln(u)$.

Then:

$$\frac{du}{dx} = 6x^2 \quad \text{and} \quad \frac{dy}{du} = \frac{1}{u}$$

Applying the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3 + 1}$$

Now evaluate the derivative at $x = 1$:

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{6(1)^2}{2(1)^3 + 1} = \frac{6}{3} = 2$$

Misconception Analysis:

- Choice B (6) occurs if one differentiates only the inner function $2x^3 + 1$ to get $6x^2$ and evaluates it at $x = 1$, neglecting the outer logarithmic derivative.
- Choice C ($\frac{2}{3}$) arises from an incorrect derivative of the inner polynomial, such as differentiating $2x^3$ incorrectly as $2x$.
- Choice D ($\ln 3$) confuses the gradient of the tangent with the y-coordinate of the point on the curve, $y(1) = \ln(2(1)^3 + 1) = \ln 3$. [M1][A1]

3 MCQ

Find the exact value of the definite integral

$$\int_0^{\ln 2} e^{2x}\,dx.$$

A $\frac{3}{2}$ ✓

B 3

C $\frac{1}{2}$

D $\frac{7}{2}$

WORKED SOLUTION & VERIFICATION

To evaluate the definite integral:

1. Find the antiderivative using the standard standard result $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$:

$$\int e^{2x}\,dx = \frac{1}{2}e^{2x}$$

2. Apply the limits from $x = 0$ to $x = \ln 2$:

$$\left[\frac{1}{2}e^{2x}\right]_0^{\ln 2} = \frac{1}{2}e^{2\ln 2} - \frac{1}{2}e^{2(0)}$$

3. Simplify the exponential terms using the power law of logarithms, $2\ln 2 = \ln(2^2) = \ln 4$:

$$e^{2\ln 2} = e^{\ln 4} = 4$$

and

$$e^0 = 1$$

4. Compute the final value:

$$\frac{1}{2}(4) - \frac{1}{2}(1) = 2 - \frac{1}{2} = \frac{3}{2}$$

Common Errors:

- Omitting the factor of $\frac{1}{2}$ gives $4 - 1 = 3$ (Choice B).
- Forgetting that $2\ln 2 = \ln 4$ and incorrectly setting $e^{2\ln 2} = 2$ gives $\frac{1}{2}(2 - 1) = \frac{1}{2}$ (Choice C). [M1][A1]

4 MCQ

The rational function f is defined for all real $x \notin \{-1, 2\}$ by

$$f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$$

where k is a constant. The partial fraction decomposition of $f(x)$ takes the form

$$A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$$

where A, B, C, and D are real constants.
Given that $C = 2$, find the value of k.

A -3 ✓

B 1

C 3

D 24

WORKED SOLUTION & VERIFICATION

Step 1: Determine the polynomial quotient A.
The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives:

$$(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$$

Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient:

$$A = \frac{1}{1} = 1$$

Step 2: Set up the algebraic identity.
Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$:

$$x^3 + kx + 7 \equiv A(x+1)(x-2)^2 + B(x-2)^2 + C(x+1)(x-2) + D(x+1)$$

Step 3: Relate B and C by comparing coefficients of x^2.
On the left-hand side, the coefficient of x^2 is 0.
On the right-hand side, collecting the x^2 terms:
- $A(x+1)(x-2)^2 = A(x^3 - 3x^2 + 4) \implies -3A$
- $B(x-2)^2 = B(x^2 - 4x + 4) \implies +B$
- $C(x+1)(x-2) = C(x^2 - x - 2) \implies +C$
- $D(x+1) \implies 0$

Therefore:

$$0 = -3A + B + C$$

Substitute $A = 1$ and the given value $C = 2$:

$$-3(1) + B + 2 = 0 \implies B = 1$$

Step 4: Solve for k by substituting $x = -1$.
Setting $x = -1$ eliminates the terms containing $(x+1)$:

$$(-1)^3 + k(-1) + 7 = B(-1-2)^2$$

$$-1 - k + 7 = B(-3)^2$$

$$6 - k = 9B$$

Since $B = 1$:

$$6 - k = 9 \implies k = -3$$

Common Algebraic Traps:
- $k = 24$: Omits the quotient A (treating an improper fraction as proper), giving $B + C = 0 \implies B = -2$, leading to $6 - k = -18 \implies k = 24$.
- $k = 1$: Substitutes $x = 1$ instead of the root $x = -1$, obtaining $1 + k + 7 = 9 \implies k = 1$.
- $k = 3$: Makes a sign error when rearranging $6 - k = 9$. [M1][A1]

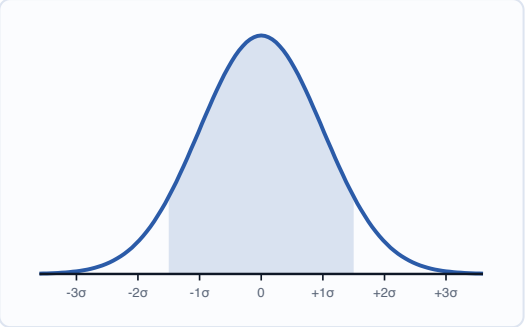
5 MCQ

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$.

The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$.

Using the standard normal critical value $z_{0.05} = 1.645$:

1. Find the value of σ.
2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.



A 43

B 11

C 22

D **44** ✓

WORKED SOLUTION & VERIFICATION

We solve the problem in two stages:

Step 1: Determine σ

Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$.

The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$:

$$\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$$

$$\frac{0.658}{\frac{\sigma}{5}} = 1.645 \implies \frac{3.29}{\sigma} = 1.645 \implies \sigma = \frac{3.29}{1.645} = 2.0 \text{ mm}$$

Step 2: Determine the minimum sample size m

For a new sample of size m, the rejection rule at the 5% level requires rejecting H_0 if $\bar{X} \geq c$, where:

$$c = 20.0 + 1.645\left(\frac{2.0}{\sqrt{m}}\right) = 20.0 + \frac{3.29}{\sqrt{m}}$$

A Type II error occurs when H_0 is not rejected (i.e. $\bar{X} < c$) given that the true mean is $\mu = 21.0$.

We require:

$$P(\bar{X} < c \mid \mu = 21.0) \leq 0.05$$

Standardising under $\mu = 21.0$:

$$\frac{c - 21.0}{\frac{2.0}{\sqrt{m}}} \leq -1.645 \implies c \leq 21.0 - 1.645\left(\frac{2.0}{\sqrt{m}}\right) = 21.0 - \frac{3.29}{\sqrt{m}}$$

Equating both conditions for c:

$$20.0 + \frac{3.29}{\sqrt{m}} \leq 21.0 - \frac{3.29}{\sqrt{m}}$$

$$\frac{6.58}{\sqrt{m}} \leq 1.0 \implies \sqrt{m} \geq 6.58$$

$$m \geq 6.58^2 = 43.2964$$

Since m must be an integer, the minimum sample size is $m = 44$.

Common traps:

- Rounding down to 43 yields a Type II error probability greater than 0.05.
- Omitting the calculation of σ and erroneously assuming $\sigma = 1$ yields $m = 11$.
- Omitting the critical value shift from one of the two tails leads to $m = 22$. [M1][A1]

6 MCQ

The complex number z satisfies the equation

$$z(2 - \mathrm{i}) = 7 + 4\mathrm{i}.$$

Express z in the form $x + \mathrm{i}y$, where x and y are real.

A $2 + 3\mathrm{i}$ ✓

B $\frac{18}{5} + \frac{1}{5}\mathrm{i}$

C $2 - 3\mathrm{i}$

D $\frac{10}{3} + 5\mathrm{i}$

WORKED SOLUTION & VERIFICATION

To find z, rearrange the given equation:

$$z = \frac{7 + 4\mathrm{i}}{2 - \mathrm{i}}$$

Multiply the numerator and the denominator by the complex conjugate of the denominator, which is $2 + \mathrm{i}$:

$$z = \frac{(7 + 4\mathrm{i})(2 + \mathrm{i})}{(2 - \mathrm{i})(2 + \mathrm{i})}$$

Expand the denominator:

$$(2 - \mathrm{i})(2 + \mathrm{i}) = 2^2 - \mathrm{i}^2 = 4 - (-1) = 5$$

Expand the numerator:

$$(7 + 4\mathrm{i})(2 + \mathrm{i}) = 14 + 7\mathrm{i} + 8\mathrm{i} + 4\mathrm{i}^2 = 14 + 15\mathrm{i} - 4 = 10 + 15\mathrm{i}$$

Divide each term by the denominator:

$$z = \frac{10 + 15\mathrm{i}}{5} = 2 + 3\mathrm{i}$$

Common Pitfalls:

- Choice B occurs from mistakenly multiplying by $2 - \mathrm{i}$ instead of the complex conjugate $2 + \mathrm{i}$.
- Choice D arises from incorrectly evaluating the denominator as $2^2 - 1 = 3$ rather than $4 - (-1) = 5$. [M1][A1]

7 MCQ

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \leq x \leq 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \leq X \leq 2)$.

- **A** $\frac{7}{8}$ ✓
- **B** $\frac{1}{8}$
- **C** $\frac{3}{4}$
- **D** $\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

To find the probability $\mathrm{P}(1 \leq X \leq 2)$, we integrate the probability density function $f(x)$ over the interval $[1, 2]$:

$$\mathrm{P}(1 \leq X \leq 2) = \int_1^2 \frac{3}{8}x^2 \, \mathrm{d}x$$

Find the antiderivative:

$$\int \frac{3}{8}x^2 \, \mathrm{d}x = \frac{3}{8} \cdot \frac{x^3}{3} = \frac{x^3}{8}$$

Evaluate between the limits $x = 1$ and $x = 2$:

$$\left[\frac{x^3}{8}\right]_1^2 = \frac{2^3}{8} - \frac{1^3}{8} = \frac{8}{8} - \frac{1}{8} = \frac{7}{8}$$

Common errors include calculating the lower tail $\mathrm{P}(0 \leq X \leq 1) = \frac{1}{8}$, or mistakenly substituting values directly into $f(x)$ instead of integrating. [M1][A1]

8 MCQ

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

- **A** 90\%
- **B** 50\%
- **C** 10\% ✓
- **D** 0\%

WORKED SOLUTION & VERIFICATION

By definition, $P(\text{Type I error}) = \alpha = 10\%$. [M1][A1]

9 MCQ

A hypothesis test is carried out at the 2% significance level. Find the probability of a Type I error.

- **A** 0\%
- **B** 98\%
- **C** 2\% ✓
- **D** 50\%

WORKED SOLUTION & VERIFICATION

By definition, $P(\text{Type I error}) = \alpha = 2\%$. [M1][A1]

10 MCQ

In an applied problem, a hypothesis test is carried out at the 2.5% significance level. Find the probability of a Type I error.

- **A** 2.5\% ✓
- B 50\%
- C 0\%
- D 97.5\%

WORKED SOLUTION & VERIFICATION

By definition, $P(\text{Type I error}) = \alpha = 2.5\%$. [M1][A1]

11 MCQ

Suppose a hypothesis test is carried out at the 1% significance level. Find the probability of a Type I error.

- A 50\%
- B 99\%
- C 0\%
- **D** 1\% ✓

WORKED SOLUTION & VERIFICATION

By definition, $P(\text{Type I error}) = \alpha = 1\%$. [M1][A1]

12 MCQ

An observed hypothesis test is carried out at the 5% significance level. Find the probability of a Type I error.

- A 50\%
- **B** 5\% ✓
- C 95\%
- D 0\%

WORKED SOLUTION & VERIFICATION

By definition, $P(\text{Type I error}) = \alpha = 5\%$. [M1][A1]

13 MCQ

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses

$$H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$$

at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

A $z = -2.0$; **reject H_0 because** $-2.0 < -1.645$ ✓

B $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

C $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

D $z = -8.0$; reject H_0 because $-8.0 < -1.645$

WORKED SOLUTION & VERIFICATION

1. Under the null hypothesis $H_0 : \mu = 25.0$, the sample mean $\bar{X}$ of a random sample of size $n = 16$ from $\text{N}(\mu, \sigma^2)$ has standard error:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.2$$

2. The standardized test statistic z is:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{24.6 - 25.0}{0.2} = \frac{-0.4}{0.2} = -2.0$$

3. Since this is a lower-tail test at the 5% level with critical value $z_{\text{crit}} = -1.645$, the critical (rejection) region is $z < -1.645$.

4. Because $z = -2.0 < -1.645$, the test statistic falls in the critical region. Therefore, we reject H_0.

Common errors:

- Omitting $\sqrt{n}$ gives $z = \frac{24.6-25.0}{0.8} = -0.5$, which leads to incorrectly failing to reject H_0.
- Dividing by n rather than $\sqrt{n}$ gives $z = \frac{-0.4}{0.8/16} = -8.0$.
- Inverting the decision rule leads to falsely accepting H_0 despite the test statistic falling in the critical region. [M1] [A1]

14 MCQ

The continuous random variable X has probability density function $f(x)$ defined by

$$f(x) = \begin{cases} kx & 0 \le x < 2, \\ k(6-2x) & 2 \le x \le 3, \\ 0 & \text{otherwise}, \end{cases}$$

where k is a positive constant. Find the exact value of the conditional probability $\mathrm{P}(X > 1 \mid X < 2.5)$.

A $\frac{9}{11}$ ✓

B $\frac{10}{11}$

C $\frac{3}{4}$

D $\frac{9}{10}$

WORKED SOLUTION & VERIFICATION

Step 1: Determine the constant k using the total probability property $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$.

$$\int_0^2 kx\,\mathrm{d}x + \int_2^3 k(6-2x)\,\mathrm{d}x = 1$$

Evaluating each integral:

$$\int_0^2 kx\,\mathrm{d}x = k\left[\frac{x^2}{2}\right]_0^2 = 2k$$

$$\int_2^3 k(6-2x)\,\mathrm{d}x = k\left[6x - x^2\right]_2^3 = k((18-9)-(12-4)) = k(9-8) = k$$

Thus, $2k + k = 3k = 1 \implies k = \frac{1}{3}$.

Step 2: Find $\mathrm{P}(X < 2.5)$.

$$\mathrm{P}(X < 2.5) = \int_0^2 \frac{1}{3}x\,\mathrm{d}x + \int_2^{2.5} \frac{1}{3}(6-2x)\,\mathrm{d}x$$

$$= \frac{2}{3} + \frac{1}{3}\left[6x - x^2\right]_2^{2.5} = \frac{2}{3} + \frac{1}{3}((15-6.25)-8) = \frac{2}{3} + \frac{1}{3}(0.75) = \frac{2}{3} + \frac{1}{4} = \frac{11}{12}$$

Step 3: Find the numerator $\mathrm{P}(1 < X < 2.5)$.

$$\mathrm{P}(1 < X < 2.5) = \int_1^2 \frac{1}{3}x\,\mathrm{d}x + \int_2^{2.5} \frac{1}{3}(6-2x)\,\mathrm{d}x$$

$$= \frac{1}{3}\left[\frac{x^2}{2}\right]_1^2 + \frac{1}{4} = \frac{1}{3}\left(2 - \frac{1}{2}\right) + \frac{1}{4} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

Step 4: Calculate the conditional probability.

$$\mathrm{P}(X > 1 \mid X < 2.5) = \frac{\mathrm{P}(1 < X < 2.5)}{\mathrm{P}(X < 2.5)} = \frac{\frac{3}{4}}{\frac{11}{12}} = \frac{3}{4} \times \frac{12}{11} = \frac{9}{11}$$

Common misconceptions:

- Using $\mathrm{P}(X > 1)$ instead of $\mathrm{P}(1 < X < 2.5)$ in the numerator yields $\frac{10/12}{11/12} = \frac{10}{11}$.

- Omitting the conditioning entirely yields $\frac{3}{4}$.
- Inverting the condition yields $\mathrm{P}(X < 2.5 \mid X > 1) = \frac{3/4}{5/6} = \frac{9}{10}$. [M1][A1]

15 MCQ

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour.
Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

A $\frac{2187}{8}\mathrm{e}^{-9} \approx 0.0337$

B $54\mathrm{e}^{-6} \approx 0.134$ ✓

C $\frac{625}{24}\mathrm{e}^{-5} \approx 0.175$

D $\frac{2}{3}\mathrm{e}^{-2} \approx 0.0902$

WORKED SOLUTION & VERIFICATION

1. **Find the parameter λ for a 1-hour period:**
Let X be the number of calls received in 1 hour. Then $X \sim \mathrm{Po}(\lambda)$.
The probability mass function is $\mathrm{P}(X = x) = \frac{\lambda^x \mathrm{e}^{-\lambda}}{x!}$.
We are given:

$$\mathrm{P}(X = 1) = 1.5\,\mathrm{P}(X = 3)$$

Substitute the formula for each probability:

$$\lambda \mathrm{e}^{-\lambda} = \frac{3}{2}\left(\frac{\lambda^3 \mathrm{e}^{-\lambda}}{3!}\right) = \frac{3}{2} \cdot \frac{\lambda^3 \mathrm{e}^{-\lambda}}{6} = \frac{\lambda^3 \mathrm{e}^{-\lambda}}{4}$$

Since $\lambda > 0$ and $\mathrm{e}^{-\lambda} > 0$, divide both sides by $\lambda \mathrm{e}^{-\lambda}$:

$$1 = \frac{\lambda^2}{4} \implies \lambda^2 = 4 \implies \lambda = 2$$

2. **Scale the parameter to the 3-hour period:**
For a Poisson process, the mean number of occurrences scales linearly with the duration of the time interval. For a 3-hour period, the rate parameter is:

$$\mu = 3\lambda = 3(2) = 6$$

Let Y be the number of calls received in a 3-hour period, so $Y \sim \mathrm{Po}(6)$.
3. **Calculate $\mathrm{P}(Y = 4)$:**

$$\mathrm{P}(Y = 4) = \frac{6^4 \mathrm{e}^{-6}}{4!} = \frac{1296\mathrm{e}^{-6}}{24} = 54\mathrm{e}^{-6} \approx 0.134$$

Common Misconceptions / Traps:
- Choice B ($\frac{2}{3}\mathrm{e}^{-2}$): Fails to adjust the parameter for the 3-hour period, mistakenly evaluating $\mathrm{P}(X = 4)$ with $\lambda = 2$.
- Choice C ($\frac{2187}{8}\mathrm{e}^{-9}$): Inverts the given ratio by solving $\mathrm{P}(X = 3) = 1.5\,\mathrm{P}(X = 1)$, which yields $\lambda = 3$ and $\mu = 9$.
- Choice D ($\frac{625}{24}\mathrm{e}^{-5}$): Adds the time length to the hourly rate ($\mu = 2 + 3 = 5$) instead of multiplying. [M1][A1]

16 MCQ

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2-X)$ m, where X is a continuous random variable with probability density function

$$f(x)=\begin{cases}\frac{3}{8}x^2, & 0\le x\le 2,\\ 0, & \text{otherwise.}\end{cases}$$

Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

A $\frac{37}{351}$

B $\frac{127}{640}$

C $\frac{74}{351}$

D $\frac{127}{351}$ ✓

WORKED SOLUTION & VERIFICATION

Let A denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side.

1. **Determine the condition for $A\ge\frac{3}{4}$:**

The area is given by $A=X(2-X)=2X-X^2$.

$$2X-X^2\ge\frac{3}{4}\iff X^2-2X+\frac{3}{4}\le 0\iff (X-\frac{1}{2})(X-\frac{3}{2})\le 0$$

Thus, $A\ge\frac{3}{4}$ corresponds to the interval $E_1=[\frac{1}{2},\frac{3}{2}]$.

Using the cumulative distribution function $F(x)=\int_0^x\frac{3}{8}t^2\,dt=\frac{x^3}{8}$ for $0\le x\le 2$:

$$P(E_1)=F(\frac{3}{2})-F(\frac{1}{2})=\frac{(\frac{3}{2})^3-(\frac{1}{2})^3}{8}=\frac{\frac{27}{8}-\frac{1}{8}}{8}=\frac{26}{64}=\frac{13}{32}$$

2. **Determine the condition for $L_{\min}\le\frac{2}{3}$:**

The shorter side length is $L_{\min}=\min(X,2-X)$.

$$\min(X,2-X)\le\frac{2}{3}\iff X\le\frac{2}{3}\quad\text{or}\quad 2-X\le\frac{2}{3}\iff X\ge\frac{4}{3}$$

Thus, $E_2=[0,\frac{2}{3}]\cup[\frac{4}{3},2]$.

3. **Find the probability of the intersection $E_1\cap E_2$:**

$$E_1\cap E_2=[\frac{1}{2},\frac{2}{3}]\cup[\frac{4}{3},\frac{3}{2}]$$

Calculating the probabilities for these two disjoint intervals:

$$P([\frac{1}{2},\frac{2}{3}])=\frac{(\frac{2}{3})^3-(\frac{1}{2})^3}{8}=\frac{\frac{8}{27}-\frac{1}{8}}{8}=\frac{37}{1728}$$

$$P([\frac{4}{3},\frac{3}{2}])=\frac{(\frac{3}{2})^3-(\frac{4}{3})^3}{8}=\frac{\frac{27}{8}-\frac{64}{27}}{8}=\frac{217}{1728}$$

$$P(E_1\cap E_2)=\frac{37+217}{1728}=\frac{254}{1728}=\frac{127}{864}$$

4. **Calculate the conditional probability:**

$$P(E_2\mid E_1)=\frac{P(E_1\cap E_2)}{P(E_1)}=\frac{\frac{127}{864}}{\frac{13}{32}}=\frac{127}{864}\times\frac{32}{13}=\frac{127}{27\times 13}=\frac{127}{351}$$

Common Misconceptions:

- Assuming the distribution is symmetric about $x = 1$ leads to doubling the lower interval's probability $2 \times \frac{37}{1728} / \frac{13}{32} = \frac{74}{351}$.
- Considering only the lower interval $X \leq \frac{2}{3}$ yields $\frac{37}{351}$.
- Inverting the condition to calculate $P(E_1 \mid E_2)$ gives $\frac{127}{640}$. [M1][A1]

17 MCQ

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests:

$$H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$$

The two-stage procedure operates as follows:
- **Stage 1:** A 1-metre sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2.
- **Stage 2:** A second, independent 1-metre sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted.

Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

A $16\mathrm{e}^{-4} \approx 0.293$

B $4\mathrm{e}^{-2} \approx 0.541$

C $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

D $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$ ✓

WORKED SOLUTION & VERIFICATION

A Type II error occurs when H_0 is accepted despite the true parameter value being $\lambda = 2$ (i.e. H_1 is true).
Under $\lambda = 2$, the independent counts $X_1, X_2 \sim \mathrm{Po}(2)$. The probability mass function is:

$$P(X = k) = \frac{2^k \mathrm{e}^{-2}}{k!}$$

Specifically:

$$P(X = 0) = \mathrm{e}^{-2}, \quad P(X = 1) = 2\mathrm{e}^{-2}, \quad P(X = 2) = 2\mathrm{e}^{-2}$$

H_0 can be accepted in two mutually exclusive ways:
1. **In Stage 1:**
H_0 is accepted immediately if $X_1 = 0$:

$$P(\text{Accept in Stage 1}) = P(X_1 = 0) = \mathrm{e}^{-2}$$

2. **In Stage 2:**
The test enters Stage 2 if $X_1 \in \{1, 2\}$. H_0 is then accepted if $X_1 + X_2 \leq 3$:
- If $X_1 = 1$, H_0 is accepted if $X_2 \leq 2$:

$$P(X_2 \leq 2) = P(X_2 = 0) + P(X_2 = 1) + P(X_2 = 2) = \mathrm{e}^{-2} + 2\mathrm{e}^{-2} + 2\mathrm{e}^{-2} = 5\mathrm{e}^{-2}$$

$$P(X_1 = 1 \cap X_2 \leq 2) = (2\mathrm{e}^{-2})(5\mathrm{e}^{-2}) = 10\mathrm{e}^{-4}$$

- If $X_1 = 2$, H_0 is accepted if $X_2 \leq 1$:

$$P(X_2 \leq 1) = P(X_2 = 0) + P(X_2 = 1) = \mathrm{e}^{-2} + 2\mathrm{e}^{-2} = 3\mathrm{e}^{-2}$$

$$P(X_1 = 2 \cap X_2 \leq 1) = (2\mathrm{e}^{-2})(3\mathrm{e}^{-2}) = 6\mathrm{e}^{-4}$$

Combining these cases:

$$P(\text{Accept in Stage 2}) = 10\mathrm{e}^{-4} + 6\mathrm{e}^{-4} = 16\mathrm{e}^{-4}$$

Therefore, the overall probability of a Type II error is:

$$P(\text{Type II error}) = \mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.135335 + 0.293050 = 0.428385 \approx 0.428$$

Common Misconceptions:
- Choice B ($16e^{-4}$) accounts only for accepting H_0 at Stage 2 and omits immediate acceptance at Stage 1 ($X_1 = 0$).
- Choice C ($4e^{-2}$) computes the conditional probability $P(\text{Accept } H_0 \mid \text{Stage 2 is reached}) = \frac{16e^{-4}}{4e^{-2}} = 4e^{-2}$ instead of the unconditional probability of a Type II error.
- Choice D computes the probability of a Type I error under H_0 (where $\lambda = 1$). [M1][A1]

18 MCQ

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

A $\frac{8}{27}$

B $\frac{7}{8}$

C $\frac{19}{27}$ ✓

D $\frac{2}{3}$

WORKED SOLUTION & VERIFICATION

Let $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$.
The conditional distribution of X given $X + Y = n$ is binomial $\mathrm{B}\left(n, \frac{\lambda}{\lambda+2\lambda}\right) = \mathrm{B}\left(n, \frac{1}{3}\right)$.
Here, $n = 3$, so $X \mid (X + Y = 3) \sim \mathrm{B}\left(3, \frac{1}{3}\right)$.
We want to find:

$$\mathrm{P}(X \geq 1 \mid X + Y = 3) = 1 - \mathrm{P}(X = 0 \mid X + Y = 3)$$

Using the conditional formula directly:

$$\mathrm{P}(X = 0 \mid X + Y = 3) = \frac{\mathrm{P}(X = 0 \cap Y = 3)}{\mathrm{P}(X + Y = 3)} = \frac{(e^{-\lambda})\left(\frac{(2\lambda)^3 e^{-2\lambda}}{3!}\right)}{\frac{(3\lambda)^3 e^{-3\lambda}}{3!}} = \frac{2^3}{3^3} = \frac{8}{27}$$

Therefore:

$$\mathrm{P}(X \geq 1 \mid X + Y = 3) = 1 - \frac{8}{27} = \frac{19}{27}$$

Common errors:
- Calculating $\mathrm{P}(X = 0 \mid X + Y = 3) = \frac{8}{27}$ and forgetting to subtract from 1.
- Assuming equal lengths, which gives $1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$.
- Taking the proportion of length $\frac{1}{1+2} = \frac{1}{3}$ and subtracting from 1 to obtain $\frac{2}{3}$. [M1][A1]

19 MCQ

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

- **A** 0.980 ✓
- **B** 0.900
- **C** 0.810
- **D** 0.990

WORKED SOLUTION & VERIFICATION

1. **Margin of error for each confidence interval:**

For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width:

$$E = 1.645\frac{\sigma}{\sqrt{n}}$$

The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$.

2. **Condition for interval overlap:**

Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths:

$$|\bar{X}_1 - \bar{X}_2| \leq 2E = 2(1.645)\frac{\sigma}{\sqrt{n}}$$

3. **Distribution of the difference between sample means:**

Since $\bar{X}_1 \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ and $\bar{X}_2 \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ are independent:

$$D = \bar{X}_1 - \bar{X}_2 \sim N\left(0, \frac{\sigma^2}{n} + \frac{\sigma^2}{n}\right) = N\left(0, \frac{2\sigma^2}{n}\right)$$

The standard deviation of D is $\sigma_D = \sqrt{\frac{2\sigma^2}{n}} = \frac{\sigma\sqrt{2}}{\sqrt{n}}$.

4. **Standardization and probability calculation:**

$$P(|D| \leq 2E) = P\left(|Z| \leq \frac{2(1.645)\frac{\sigma}{\sqrt{n}}}{\sqrt{2}\frac{\sigma}{\sqrt{n}}}\right) = P(|Z| \leq 1.645\sqrt{2})$$

Evaluating the boundary:

$$1.645\sqrt{2} \approx 1.645 \times 1.41421 \approx 2.326$$

Using the symmetry of the standard normal distribution:

$$P(|Z| \leq 2.326) = 2\Phi(2.326) - 1 = 2(0.9900) - 1 = 1.9800 - 1 = 0.980$$

Common Misconceptions:

- **0.900:** Mistakenly assuming the probability of overlap equals the confidence level of a single interval.
- **0.810:** Assuming the intervals overlap only when both contain μ, giving $0.90 \times 0.90 = 0.810$. While both containing μ is sufficient, intervals can also overlap without containing μ.
- **0.990:** Computing the probability that at least one interval contains μ $(1 - 0.10^2 = 0.990)$ or reading $\Phi(2.326)$ without subtracting the lower tail. [M1][A1]

20 MCQ

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} k(4-x^2) & 0 \le x \le 2, \\ 0 & \text{otherwise}, \end{cases}$$

where k is a constant.
Find the value of k.

A $\frac{3}{16}$ ✓

B $\frac{3}{8}$

C $\frac{16}{3}$

D $\frac{1}{4}$

WORKED SOLUTION & VERIFICATION

For $f(x)$ to be a valid probability density function, the total area under the curve must equal 1:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Substitute the non-zero definition of $f(x)$:

$$\int_{0}^{2} k(4-x^2)\,dx = 1$$

Factor out the constant k and integrate term by term:

$$k\left[4x - \frac{x^3}{3}\right]_0^2 = 1$$

Evaluate at the upper and lower limits:

$$k\left((4(2) - \frac{2^3}{3}) - 0\right) = 1$$

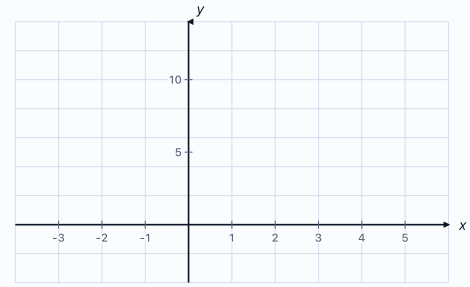
$$k\left(8 - \frac{8}{3}\right) = 1$$

$$k\left(\frac{16}{3}\right) = 1 \implies k = \frac{3}{16}$$

Common errors include:
- Integrating x^2 as x^2 or dividing by 2 instead of 3, leading to $k = \frac{1}{4}$.
- Integrating over $[-2, 2]$ instead of $[0, 2]$, which gives double the integral and results in $k = \frac{3}{8}$.
- Evaluating the integral correctly as $\frac{16}{3}$ but forgetting to invert to solve for k, yielding $\frac{16}{3}$. [M1][A1]

21 MCQ

Referring to the coordinate curve shown in the diagram, a hypothesis test at the 1% significance level gives a p-value of 6%. What is the decision?



- **A** Do not reject H_0 ✓
- **B** Test is inconclusive
- **C** Reject H_0
- **D** Increase the sample size

WORKED SOLUTION & VERIFICATION

Since the p-value $\geq \alpha$ ($6\% \geq 1\%$), we do not reject H_0. [M1][A1]

22 MCQ

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.

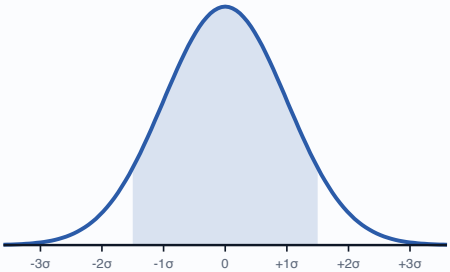
- **A** 1.96
- **B** 1.282
- **C** 2.576
- **D** **1.645** ✓

WORKED SOLUTION & VERIFICATION

The critical value for a 5% (one-tail) test is $z = 1.645$. [M1][A1]

23 MCQ

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.



- **A** 2.576
- **B** 1.645
- **C** 1.282
- **D** **1.96** ✓

WORKED SOLUTION & VERIFICATION

The critical value for a 5% (two-tail) test is $z = 1.96$. [M1][A1]

24 MCQ

A hypothesis test at the 1% significance level gives a p-value of 4%. What is the decision?

- A Reject H_0
- B Increase the sample size
- C Test is inconclusive
- D **Do not reject H_0** ✓

WORKED SOLUTION & VERIFICATION

Since the p-value $\geq \alpha$ ($4\% \geq 1\%$), we do not reject H_0. [M1][A1]

25 MCQ

A hypothesis test at the 10% significance level gives a p-value of 15%. What is the decision?

- A Increase the sample size
- B **Do not reject H_0** ✓
- C Reject H_0
- D Test is inconclusive

WORKED SOLUTION & VERIFICATION

Since the p-value $\geq \alpha$ ($15\% \geq 10\%$), we do not reject H_0. [M1][A1]